

CS & IT ENGINEERING

DATA STRUCTURE

Tree (Part-II)



JOIN TELEGRAM TO ENROLL

SHAIENDRA SIR

Binary Tree

- ✓ Binary Tree
- ✓ Representation of Binary Tree
- ✓ Types of Binary Tree
- ✓ Properties of Binary Tree
- ✓ Traversal of Binary Tree

Memory Representation of Binary Tree:-

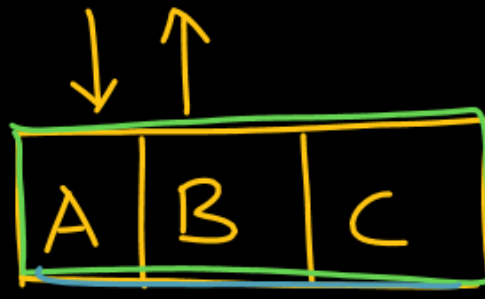
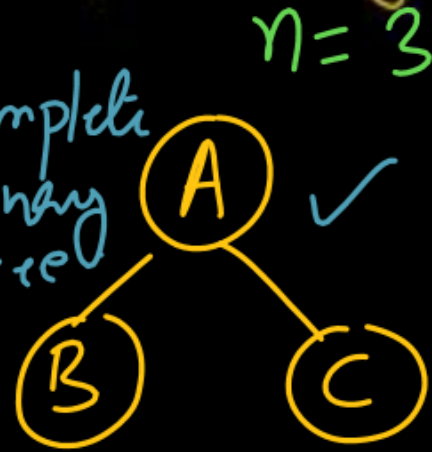
Array Representation:-

Algo:- Parent(i)
- return $\lfloor i/2 \rfloor$

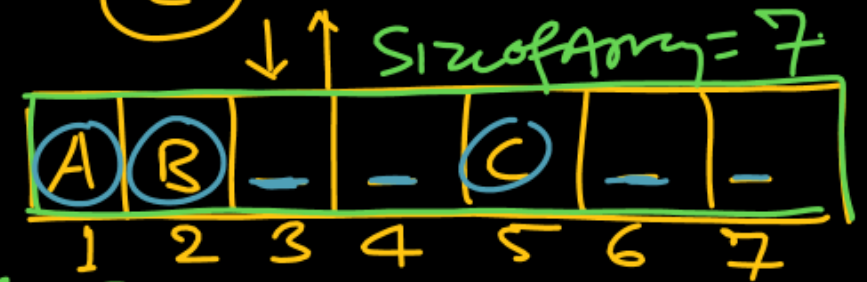
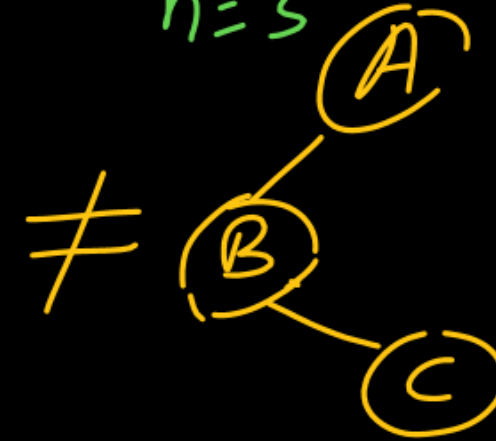
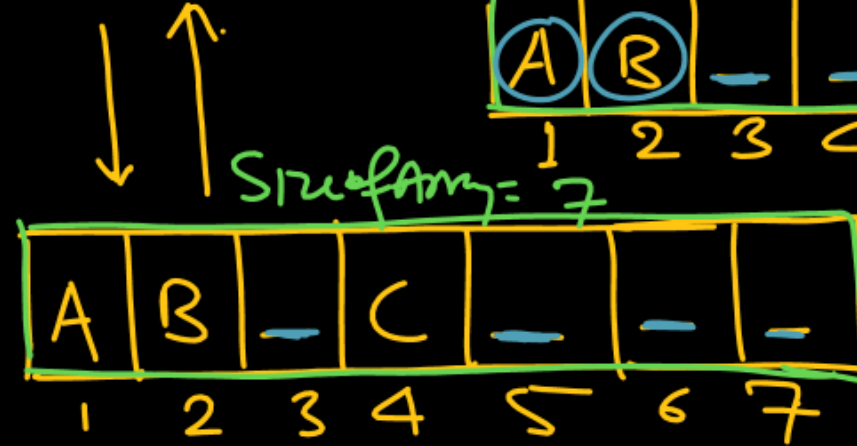
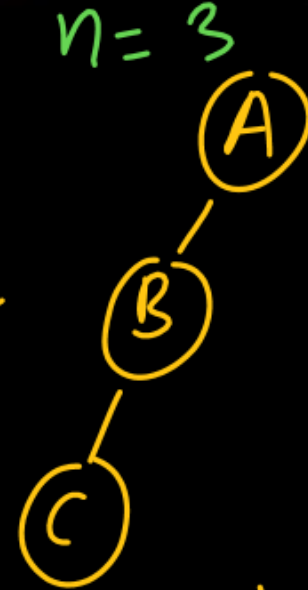
Left(i)
- return $2i$

Right(i)
- return $2i+1$

Complete
Binary
Tree



Size of Array = 3



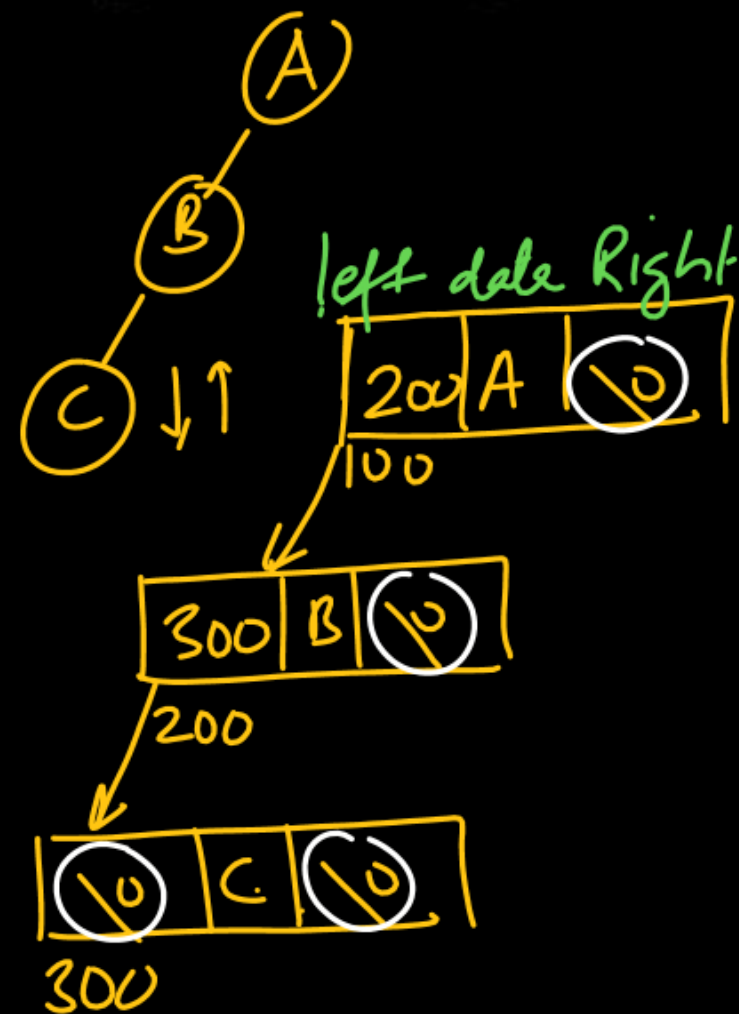
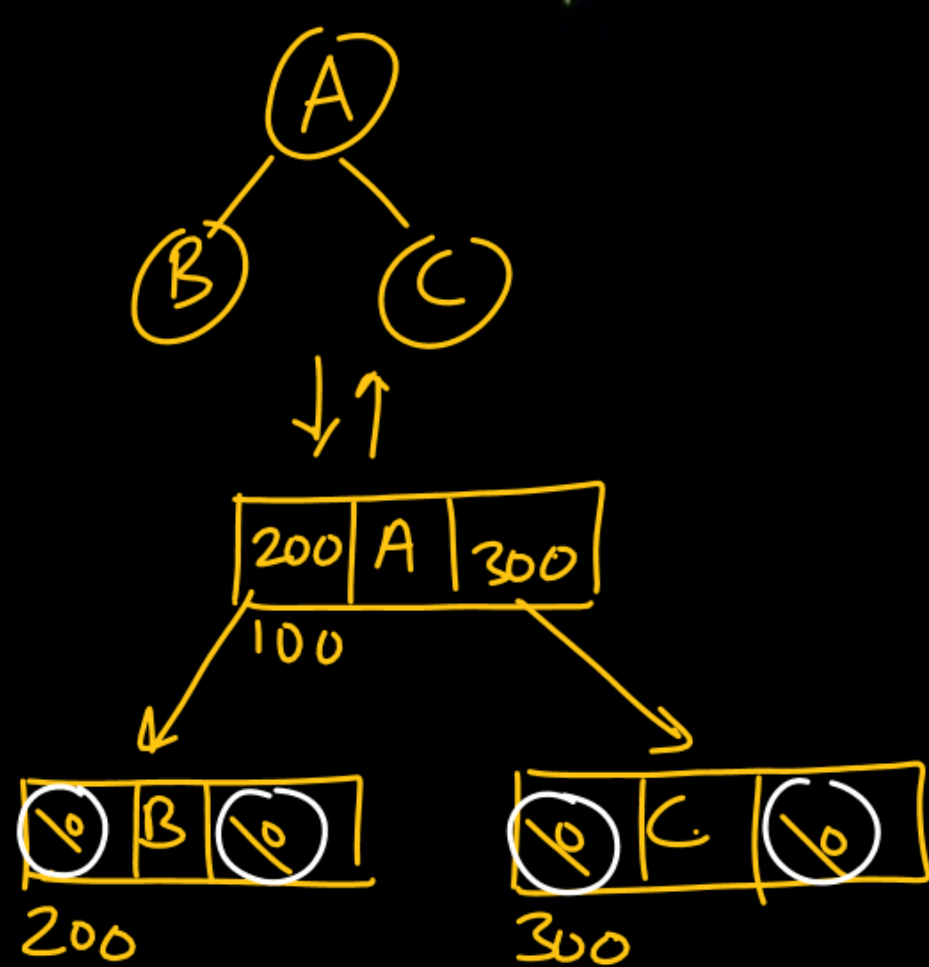
Size of Array = 7

Size of Array = 7

if n number of search keys then maximum and minimum size of Array?

Minimum Size of Array $\Rightarrow 2^n - 1$

Linked List Representation:- Data Stored in the form of Node



NOTE:- 1> If no. of keys n Then maximum and minimum number of nodes = n
2> Number of null pointers = $n+1$

node a = { NULL, 10, NULL };

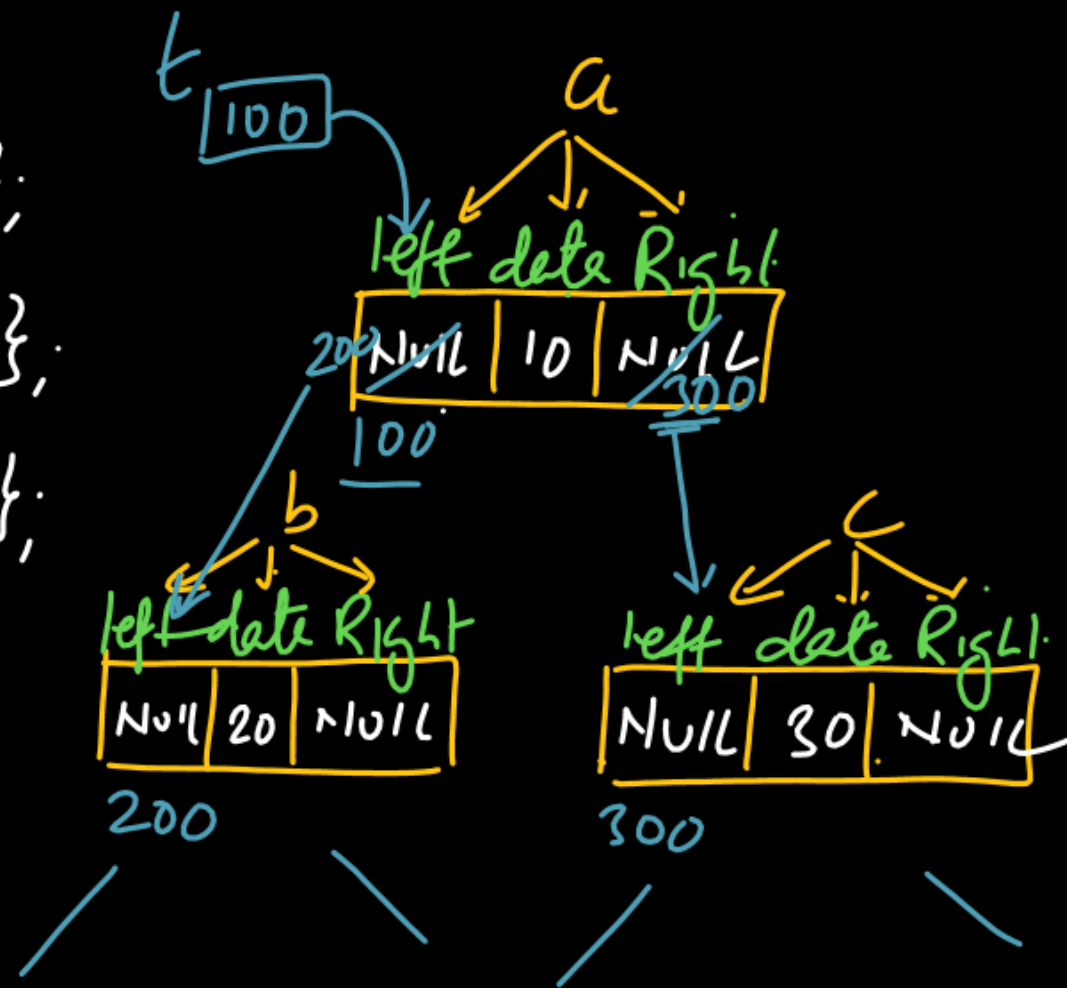
node b = { NULL, 20, NULL };

node c = { NULL, 30, NULL };

a.left = &b;

a.Right = &c;

node *t = &a;



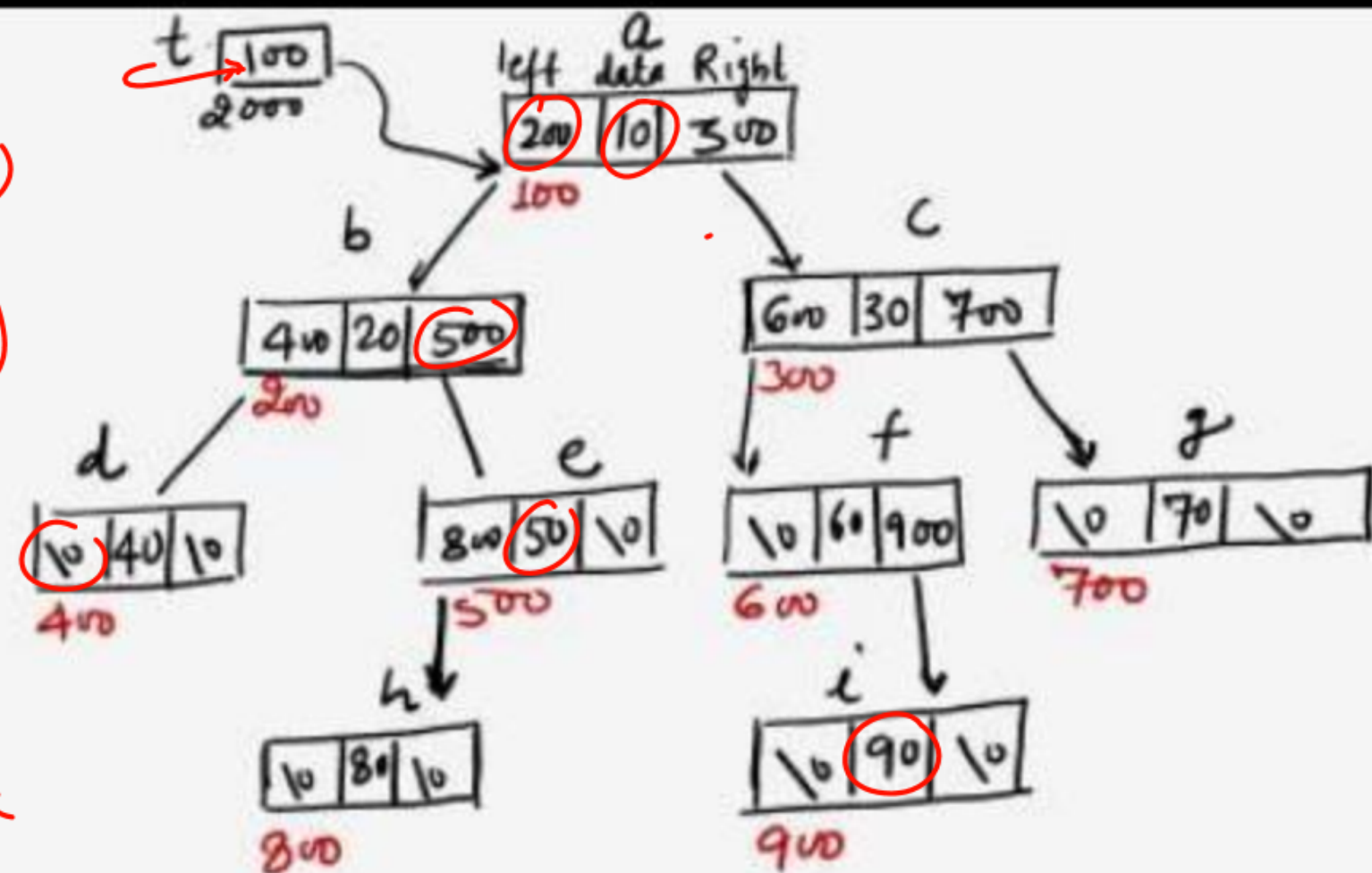
Struct BTreeNode

```
{  
    Struct BTreeNode *left;  
    int data;  
    Struct BTreeNode *Right;  
};
```

typedef BTreeNode node;

- 1) $Pf(t) = ? // 100$
- 2) $Pf(t \rightarrow \text{data}) // 10$
- 3) $Pf(t \rightarrow \text{left}) // 200$
- 4) $\text{if } (t == \text{NULL}) \text{ (or) } \text{if } (!t) // \text{Suppose This is True Then}$
- 5) $\text{if } ((t \rightarrow \text{left} == \text{NULL}) \& \& (t \rightarrow \text{right} == \text{NULL}))$
 (or)
 $\text{if } (! (t \rightarrow \text{left}) \& \& ! (t \rightarrow \text{right}))$
 { if True.
 Then
 it is
 leaf node
- 6) $(t \rightarrow \text{left})$ is given? // left subtree gives (200.)
- 7) $(t \rightarrow \text{right})$ is given? // Right subtree gives (300)
- 8) $Pf(t \rightarrow \text{left} \rightarrow \text{right} \rightarrow \text{data}) = ? // (50)$
 $200 \rightarrow 500 \rightarrow$
- 9) $Pf(t \rightarrow \text{right} \rightarrow \text{left} \rightarrow \text{right} \rightarrow \text{data}) = ? // (90)$
 $300 \rightarrow 600 \rightarrow 900 \rightarrow$
- 10) $Pf(t \rightarrow \text{left} \rightarrow \text{left} \rightarrow \text{left} \rightarrow \text{data}) = ? // \text{Error b/c NULL Pointer Dereferencing}$
 $200 \rightarrow 400 \rightarrow \text{NULL} \rightarrow$

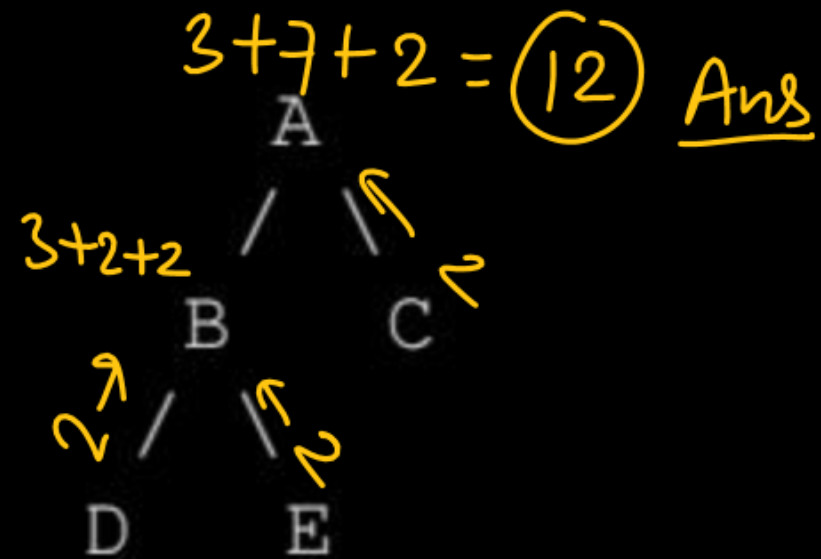
(Empty Binary Tree)



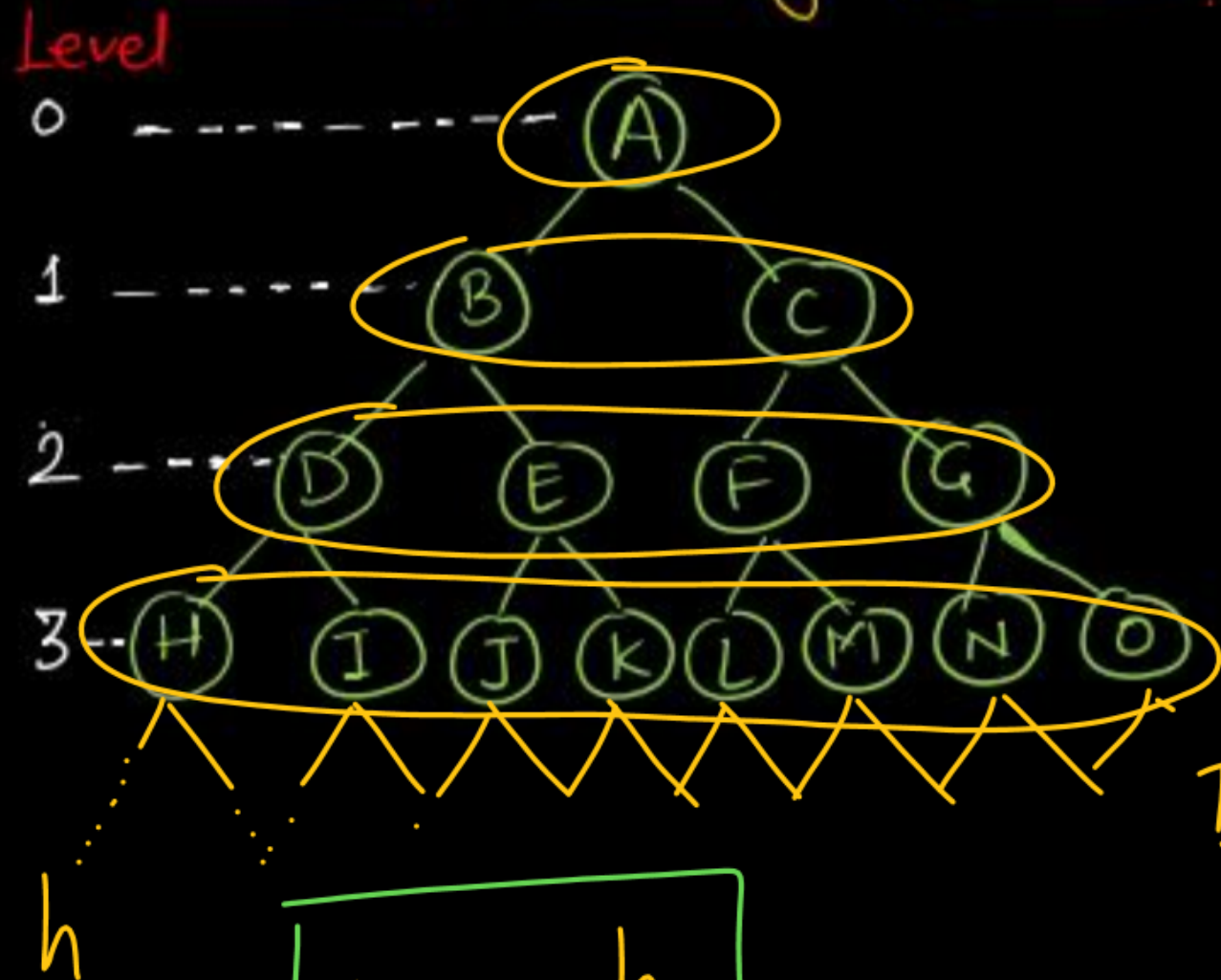
Question:

What does this function compute for a given binary tree?

```
int func(Node *root)
{
    if (root == NULL) return 0;
    else if (root->left == NULL && root->right == NULL)
        return 2;
    else
        return (3 + func(root->left) + func(root->right));
}
```



Relationship among height, Total nodes, Left nodes and Internal nodes:-



Internal nodes:

$$I = 2^0 + 2^1 + 2^2 + \dots + 2^{h-1}$$

$$I_{\max} = 2^h - 1$$

Total Nodes:

$$N = 2^0 + 2^1 + 2^2 + \dots + 2^h$$

$$N_{\max} = 2^{h+1} - 1$$

$$L_{\max} = 2^h$$

leaf nodes ↑

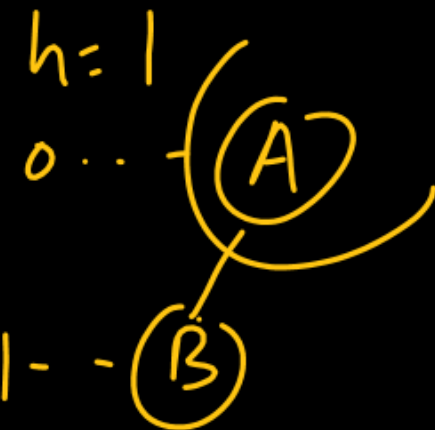
Relationship among height, Total nodes, Left nodes and Internal nodes:-

$h=0$ (A)

$$I_{min} = h$$

$$N_{min} = h + 1$$

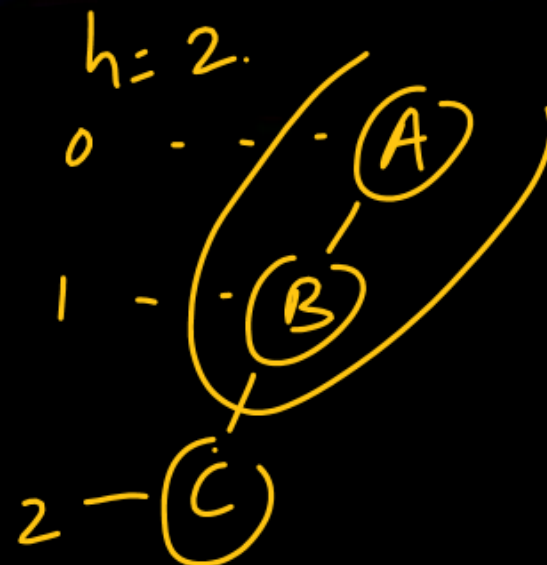
$$L_{min} = 1$$



$$I_{min} = h$$

$$N_{min} = h + 1$$

$$L_{min} = 1$$



$$I_{min} = h$$

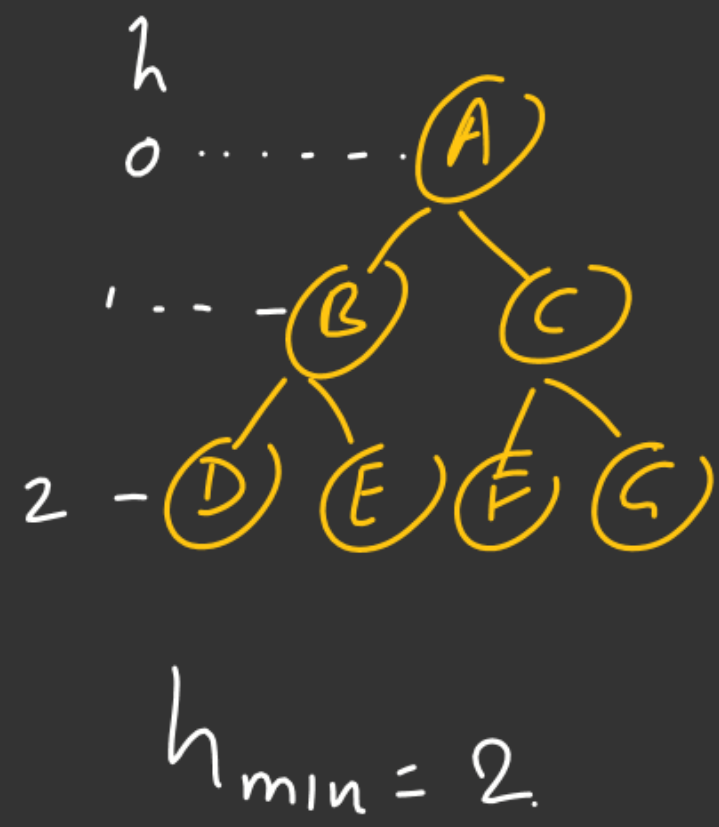
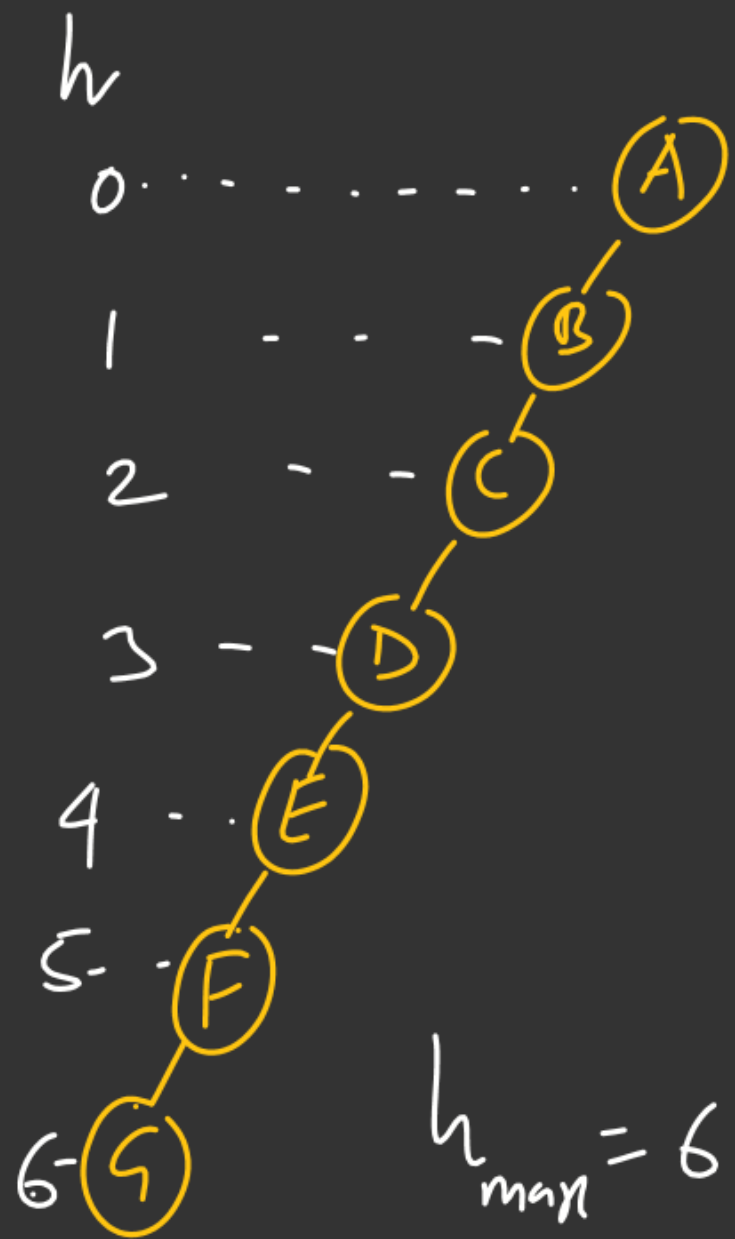
$$N_{min} = h + 1$$

$$L_{min} = 1$$

$$I_{min} = h$$

$$N_{min} = h + 1$$

$$L_{min} = 1$$



$$N = 2^{h+1} - 1$$

$$h+1 = \log_2(N+1)$$

$$h_{\min} = \log_2(N+1) - 1$$

or

$$h_{\min} = \log_2\left(\frac{N+1}{2}\right)$$

$$N = h+1$$

$$h_{\max} = N - 1$$

(A)
 $I = 0$
 $L = 1$

(A)
|
(B)
 $I = 1$
 $L = 1$

(A)
|
(B)
|
(C)
 $I = 2$
 $L = 1$

(A)
/ \
(B) (C)
 $I = 1$
 $L = 2$

(A)
/ \
(B) (C)
|
(D)
 $I = 2$
 $L = 2$

(A)
/ \
(B) (C)
/ \
(D) (E) (F) (G)
 $I = 3$
 $L = 4$

$$L \leq I + 1$$

$$h+1 \leq N \leq 2^{h+1} - 1$$

$$h \leq I \leq 2^h - 1$$

$$1 \leq L \leq 2^h$$

$$\log_2(N+1) - 1 \leq h \leq N - 1$$

Properties of Binary Tree:-

Let T be a binary tree with 15 nodes. The minimum and maximum possible heights of T are :
Note: The height of a tree with a single node is 0.

$$\log_2(N+1) - 1 \leq h \leq N - 1$$

$$3 \leq h \leq 14$$

Ans.

The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are

$$h+1 \leq N \leq 2^{h+1} - 1$$

$$6 \leq N \leq 63$$

Ans.

Let H , I , L , and N represent height, number of internal nodes, number of leaf nodes, and the total number of nodes respectively in a rooted binary tree.

Which of the following statements is/are always TRUE?

✓ A. $L \leq I + 1$

✓ B. $H + 1 \leq N \leq 2^{H+1} - 1$

✓ C. $H \leq I \leq 2^H - 1$

✗ D. $H \leq L \leq 2^{H-1}$

- For any non empty Binary Tree T, if n_0 is the number of leaf nodes (degree-0) and n_2 is the number of internal nodes (degree-2) $n_0 = n_2 + 1$

Proof:- n is total no. of nodes in Binary Tree and n_i is the no. of nodes having degree i ($0 \leq i \leq 2$)

$$n = n_0 + n_1 + n_2 \dots (i)$$

$$e = n_0 \times 0 + n_1 \times 1 + n_2 \times 2$$

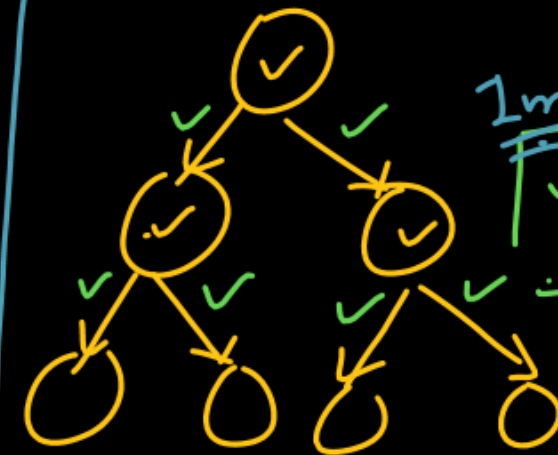
$$n-1 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2$$

$$n = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + 1 \dots (ii)$$

$$n_0 \propto n_2$$

$$n_0 \uparrow \Leftrightarrow n_2 \uparrow$$

$$n_0 \downarrow \Leftrightarrow n_2 \downarrow$$



Imp

$$e = n - 1$$

Imp

$$n_0 + n_1 + n_2 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + 1$$

$$n_0 + \cancel{n_1} + n_2 = \cancel{n_0} \times 0 + \cancel{n_1} \times 1 + n_2 \times 2 + 1$$

Imp

$$n_0 = n_2 + 1$$

Imp

$$n_2 = n_0 - 1$$

$$n_0 = 4, n_1 = 0, n_2 = 2$$

$$e = n_0 \times 0 + n_1 \times 1 + n_2 \times 2$$

$$e = 4 \times 0 + 0 \times 1 + 2 \times 2$$

$$e = 6$$

- For any non empty Binary Tree T , if n_0 is the number of leaf nodes (degree-0) and n_2 is the number of internal nodes (degree-2) $n_0 = n_2 + 1$

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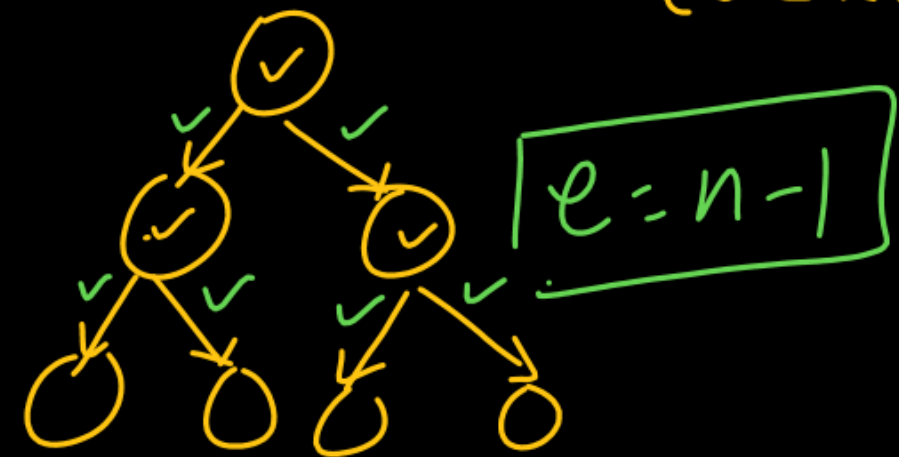
$$n-1 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2$$

Binary Tree $n = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + 1 \dots (ii)$

$$n_0 + n_1 + n_2 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + 1$$

Similarly for Ternary Tree:

$$n_0 + n_1 + n_2 + n_3 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + n_3 \times 3 + 1$$



$$n_0 = 4, n_1 = 0, n_2 = 2$$

$$e = n_0 \times 0 + n_1 \times 1 + n_2 \times 2$$

$$e = 4 \times 0 + 0 \times 1 + 2 \times 2$$

$$e = 6$$

Consider a binary tree T that has 200 leaf nodes. Then, the number of nodes in T that have exactly two children are 199.

$$n_0 = n_2 + 1$$

$$n_2 = n_0 - 1$$

Question:- In a Binary Tree, the number of internal nodes of degree 1 is 5 and number of Internal node of degree 2 is 10 the number of Leaf nodes in the Binary tree?

$$n_0 = n_2 + 1 \Rightarrow 10 + 1 \Rightarrow \textcircled{11} \underline{\text{Ans}}$$

$$n_0 + n_1 + n_2 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + 1$$

$$n_0 + 5 + 10 = \cancel{n_0} \times 0 + 5 \times 1 + 10 \times 2 + 1$$

$$n_0 = 26 - 15 \Rightarrow \textcircled{11} \underline{\text{Ans}}$$

Question:- In a ternary tree, the number of internal nodes of degree 1, 2 and 3 are 3, 5 and 4 respectively. The number of Leafnodes in the given tree is —?

$$n_0 + n_1 + n_2 + n_3 = n_0 \times 0 + n_1 \times 1 + n_2 \times 2 + n_3 \times 3 + 1$$

$$n_0 + 3 + 5 + 4 = 3 \times 1 + 5 \times 2 + 4 \times 3 + 1$$

$$n_0 = 26 - 12 \Rightarrow \textcircled{14} \text{ Ans}$$

Number of Binary Tree:- if n number of keys or nodes then

if $n=3$

$$\text{No. of Unlabelled Binary Tree} = \frac{2^n C_n}{n+1}$$

1) Unlabelled BT = ? $\frac{6 C_3}{4} \Rightarrow \textcircled{5}$
Ans

$$\text{No. of Labelled Binary Tree} = \frac{2^n C_n}{n+1} \times n!$$

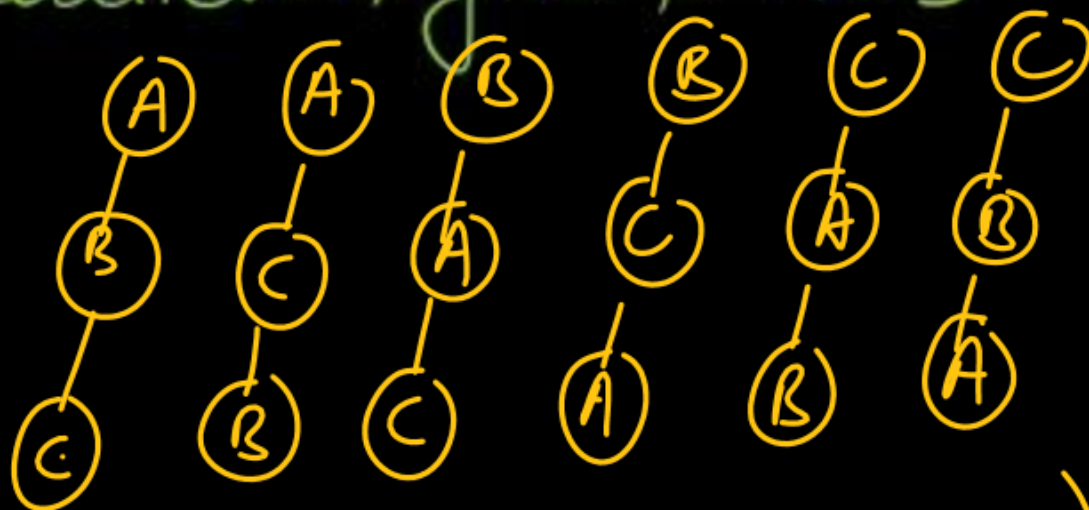
2) Labelled BT = ?
 $5 \times 3! = \textcircled{30}$
Ans

Unlabelled Binary Tree for $n=3$



Unlabelled Binary Tree.

Labelled Binary Tree for $n=3$



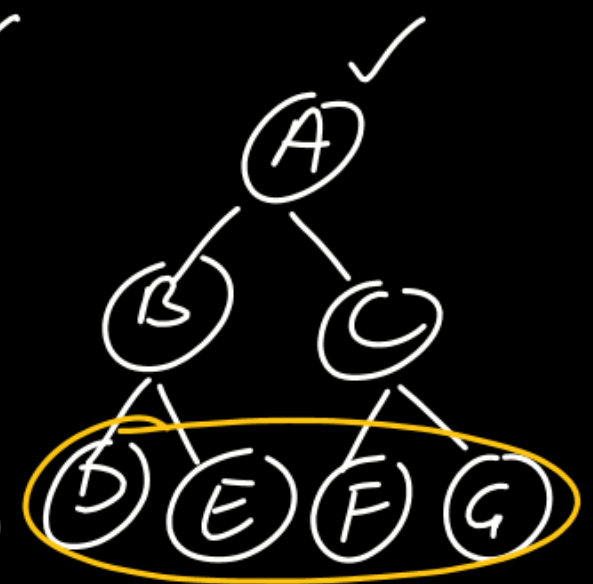
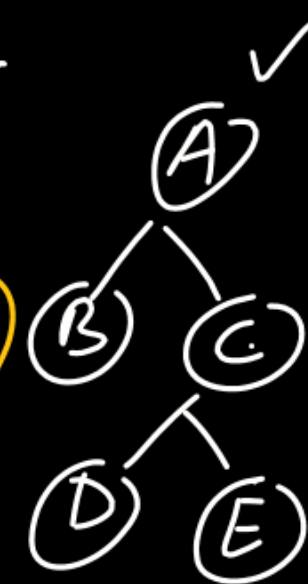
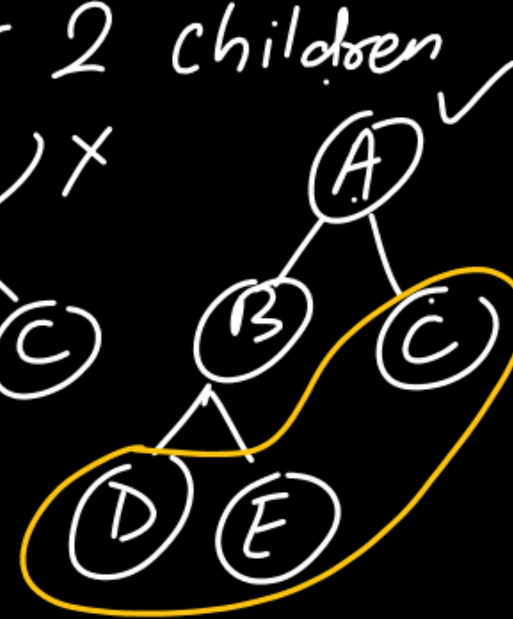
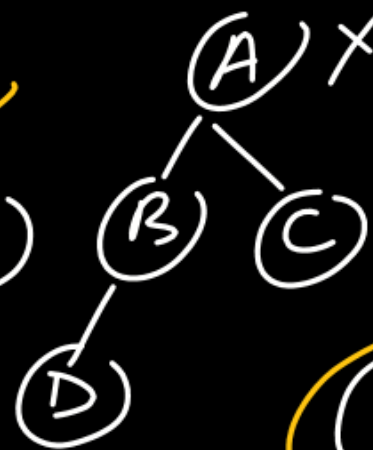
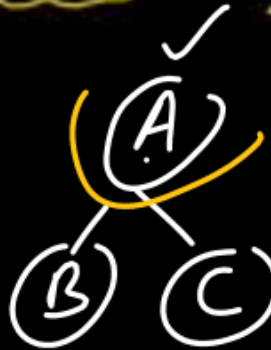
Labelled $\Rightarrow 3! \Rightarrow \textcircled{6}$

Types of Binary Tree:-

- ✓ Full / proper / Strictly Binary Tree
- ✓ Complete Binary Tree
- ✓ Perfect Binary Tree
- ✓ Degenerate Binary Tree
- ✓ Binary Search Tree (BST)
- ✓ Balanced Binary Tree
 - AVL Tree ✓
 - Red Black Tree

not in syllabus

Full/proper/Strictly Binary Tree :- ? 0 or 2 children



$$L = I + 1$$

$$N = L + I$$

$$N = 2I + 1$$

$$N = 2L - 1$$

$$I = \frac{N-1}{2}$$

Always
Odd

$$L = \frac{N+1}{2}$$

EX: $I = 15$

$N = ?$ 31

$L = ?$ 16

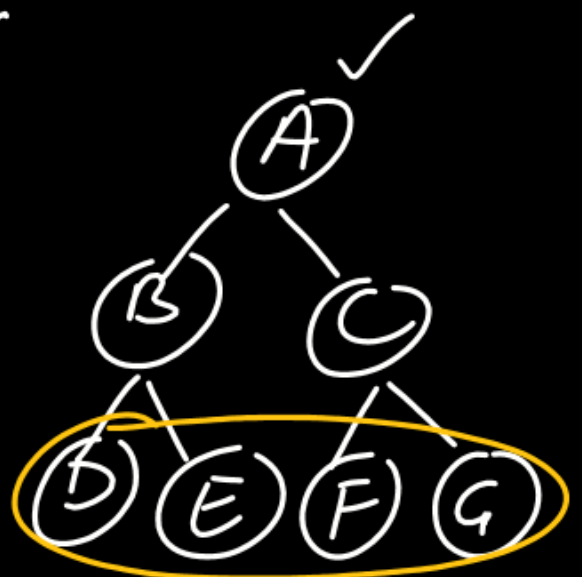
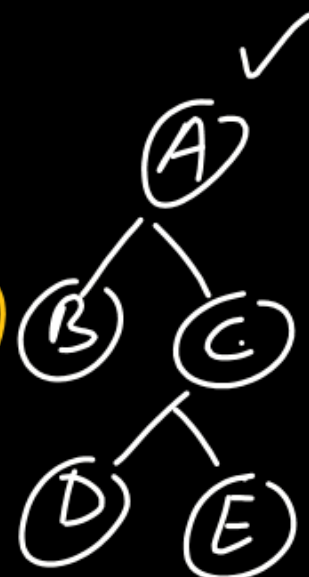
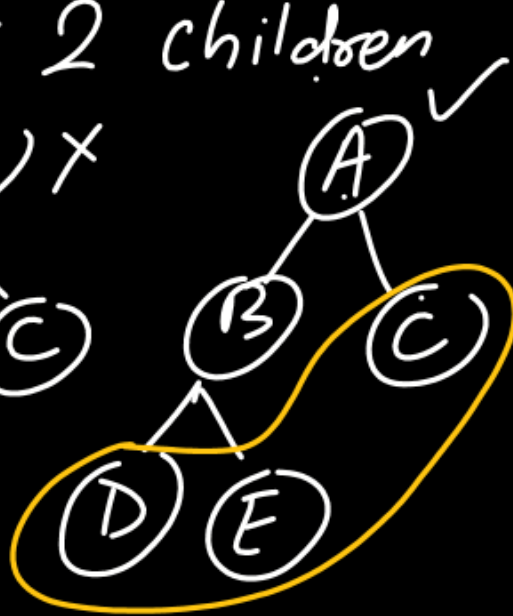
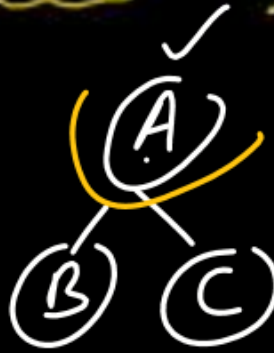
EX: $L = 32$

$N = ?$ 63

$I = ?$ 31

Full/proper/Strictly Binary Tree :- ?

0 or 2 children



$$L = I + 1$$

$$N = L + I$$

$$N = 2I + 1$$

$$N = 2L - 1$$

$$I = \frac{N-1}{2}$$

Always
Odd

$$L = \frac{N+1}{2}$$

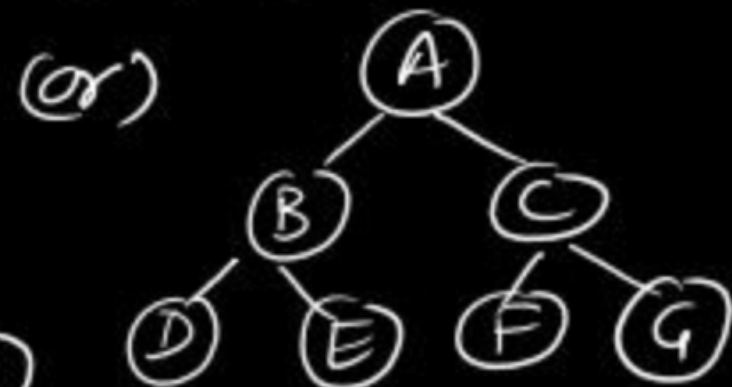
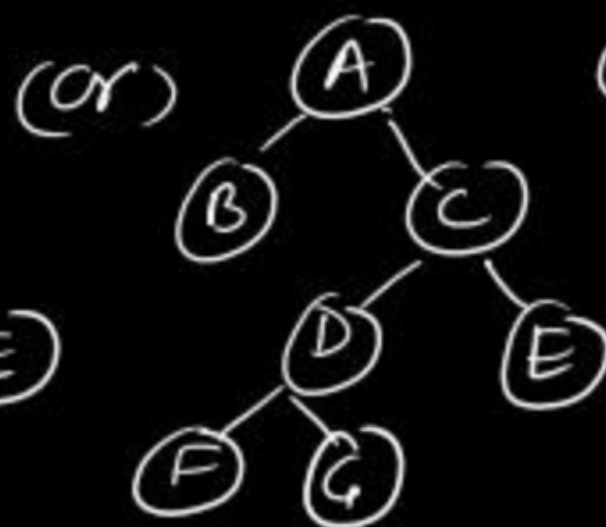
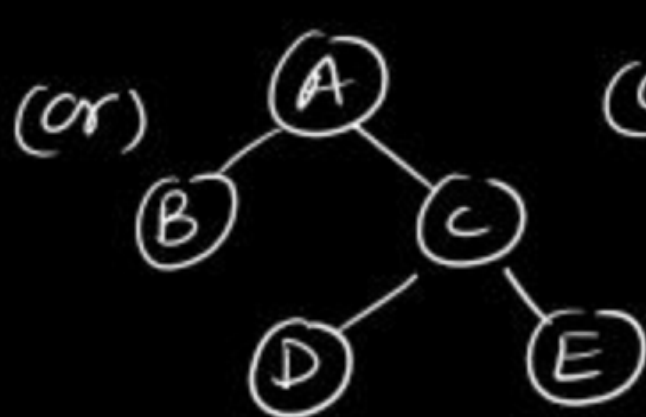
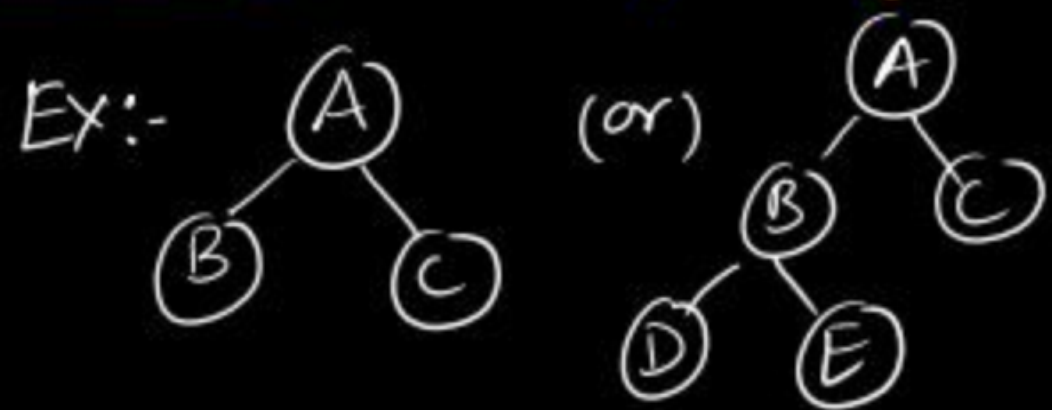
$$2^{h+1} \leq N \leq 2^{h+1} - 1$$

$$\log_2(N+1) - 1 \leq h \leq \frac{N-1}{2}$$

Ex: $N=15$, $3 \leq h \leq 7$
Ans.

Full/Proper/Strictly Binary tree:-

each node must contain either 0 or 2 children



Properties of Strictly Binary Tree:-

- The no. of Leaf node is equal to the number of internal nodes + 1
- if Leaf nodes n then total no. of nodes = $2n - 1$ (odd)
- Strictly Binary Tree have odd number of nodes
- Min no. of nodes if height h are $2h + 1$
- Max no. of nodes = $2^{h+1} - 1$
- Min height = $\log_2(n+1) - 1$
- Max height = $(n-1)/2$

$$2h+1 \leq n \leq 2^{h+1} - 1$$

↑
Min nodes

↑
Max nodes

$$\log_2(n+1) - 1 \leq h \leq \frac{n-1}{2}$$

↑
Min height

↑
Max height



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THANKYOU



GeeksforGeeks GATE